

Andrianirina Rakotoharisoa

Machine Learning Engineer / Computer Vision & Generative AI

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PROFILE

PhD graduate from Imperial College London (viva passed July 2026) with expertise in computer vision, generative AI, and applied ML engineering. Experienced designing and deploying vision pipelines, including anomaly detection, segmentation and 3D systems, on large-scale real-world sensor data. Published at NeurIPS and ICLR. Comfortable at the research frontier and in engineering pipelines: from model design and benchmarking to efficient inference and deployment. Eager to join a fast-moving team building real-time operational intelligence from sensor data.

TECHNICAL SKILLS

Engineering Practice	Parallel Computing, CI/CD, experiment tracking (hyperparameter tuning)
Languages	Python (primary), R, SQL, Shell scripting
ML Frameworks	PyTorch, Tensorflow; diffusion models, CNNs, transformers, segmentation & detection architectures
Inference & Deployment	HPC at scale (SLURM/PBS), edge deployment, cloud infrastructure
Computer Vision	Object detection & tracking, 3D reconstruction, scene understanding, multi-person tracking
Data	Multispectral & hyperspectral imagery, spatial-temporal data, time-series, video streams

RESEARCH & ENGINEERING EXPERIENCE

Visiting PhD Researcher | Computer Vision & Anomaly Detection

Sep 2025 – Dec 2025

ESA Φ-lab | Frascati, IT

- Designed and adapted ADECEES, a state-of-the-art **ensemble segmentation framework** for real-time anomaly detection in Sentinel-3 multispectral satellite imagery; accepted at NeurIPS 2025.
- Architected end-to-end **inference pipeline**: data ingestion from multispectral sensors, multi-model ensemble inference, anomaly localisation, and evaluation.
- Benchmarked novel detection and segmentation architectures on high-dimensional imagery at scale, directly analogous to transforming raw sensor feeds into operational intelligence.
- Delivered production-quality code on institutional HPC (SLURM), emphasising reproducibility and deployment readiness.

PhD Researcher | Generative AI & Deep Learning

Oct 2022 – Apr 2026

Imperial College London, Data Science Imperial | London, UK

- Adapted **diffusion models** and **super-resolution architectures** to high-dimensional geospatial sensor data, building efficient training and inference pipelines on large-scale noisy imagery.
- Released a globally-consistent daily XCO₂ dataset achieving **16× spatial resolution improvement** via a trained super-resolution network, analogous to enhancing low-resolution sensor output to high-fidelity scene understanding; published in *Remote Sensing* (2025).
- Designed and ran rigorous ML experiments and **inference workflows** on institutional HPC clusters (SLURM/PBS); optimised for reproducibility, throughput, and efficiency.
- Published peer-reviewed research at **NeurIPS** and **ICLR** workshops; lectured ML at Data Science Imperial summer schools.

System Engineer | Human-Computer Interaction & 3D Sensing

Oct 2020 – Sep 2022

Data Science Imperial, ICL | London, UK

- Led applied research on **multi-person tracking** and spatial interaction modelling using 3D sensor arrays (camera arrays, depth sensors).
- Designed and maintained scalable tracking pipelines for the Global Data Observatory, handling **real-time multi-camera video streams**.
- Managed cloud infrastructure and **security protocols** for a large-scale research facility.

MSc Research | 3D Gaze Estimation & Multi-Person Tracking
Data Science Institute, ICL | London, UK

May 2019 – Oct 2019

- Built a **3D gaze direction predictor** combining face orientation and body posture estimation in semi-open immersive environments. Hands-on 3D geometry and depth-aware CV.
- Implemented **multi-person tracking** model robust to viewpoint variation, laying early foundations in real-time vision for complex environments.

Data Scientist, Intern

Azerion | Paris, FR

May 2018 – Sep 2018

- Led a commercial gaze-tracking study; analysed time-series engagement data and delivered business KPI reports. Early experience applying CV research in a product setting.

PUBLICATIONS & CONFERENCES

Diffusion-Based Anomaly Detection for Satellite Monitoring of CO₂ Point Sources

Preprint — Mar 2026

Rakotoharisoa, A.; Cenci, S.; Arcucci, R. [doi:10.2139/ssrn.6368100](https://doi.org/10.2139/ssrn.6368100)

A High Resolution Spatially Consistent Global Dataset for CO₂ Monitoring

Remote Sensing, 2025 — May 2025

Rakotoharisoa, A.; Cenci, S.; Arcucci, R. *Remote Sens.* 2025, 17, 1617. 16× resolution improvement over source data.

ADECEES: Anomaly Detection of CO₂ Emissions via Ensemble Segmentation

NeurIPS 2025: Tackling Climate Change with ML Workshop — Dec 2025

Global High Resolution CO₂ Monitoring using Super Resolution

ICLR 2024: Tackling Climate Change with ML Workshop — May 2024

High Resolution Greenhouse Gas Monitoring using Super Resolution (Poster)

ECMWF-ESA Workshop: ML for Earth System Observation and Prediction — May 2024

EDUCATION

PhD | Machine Learning & Data Science

Imperial College London | London, UK

2022 – 2026

- Thesis: *Developing AI for Good: Increasing the Potential of Remote Sensing Data for Climate Change Mitigation*
- Supervisors: Dr Rossella Arcucci (Primary), Dr Simone Cenci (Secondary)
- Affiliated with the Data Science Institute; teaching assistant for ML, Data Science, and Big Data Assimilation courses

Diplôme d'Ingénieur, First Class

École Centrale de Nantes | Nantes, FR

2016 – 2020

- Specialisation: Applied Mathematics & Computing. Stochastic processes, Bayesian inference, deep learning, linear algebra
- Research project: ML on accelerometer data for pedestrian movement classification

TRAINING & CERTIFICATIONS

- **Generative Modeling Summer School (GeMSS)**, DTU Copenhagen, 2023: 25.5 hrs of lectures on diffusion models, VAEs, normalising flows, autoregressive models, and causal inference; hands-on sessions and poster presentations.
- **TensorFlow 2 for Deep Learning Specialisation**: probabilistic neural networks and advanced TF workflows.

LANGUAGES

French (native) • English (bilingual) • German (professional working proficiency)